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GitHub Link	https://github.com/kamesh452/MLA0405.git

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 2

Mock Interview

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 5: Autoencoders, Undercomplete Autoencoder, Regularized Autoencoder, Sparse Autoencoder, Denoising Autoencoder, Stochastic Encoders and Decoders, Contractive Autoencoder, Deep Belief Networks (DBN), Boltzmann Machines (BM), Deep Boltzmann Machines (DBM), Generative Adversarial Networks (GANs), Generator, Discriminator, GAN Applications.

Course Outcome Weightage: 35%
Assessment Tool Weightage: 20%

Duration: 90 minutes
Mark : 25

Code of Conduct:

IN.Kameshwar reddy(192525272)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *N.Kameshwar reddy* **Name of the student:** N.Kameshwar reddy

Criteria	Subject Knowledge and Conceptual Understanding (3)	Technical Accuracy and Application of Concepts (2.5)	Problem-Solving and Analytical Thinking (2)	Communication Skills and Clarity of Explanation (1.5)	Confidence, Presentation and Professional Behaviour (1)	Total (10)
Weightage	30 %	25%	20%	15 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
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Q8						
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Q22						
Q23						
Q24						
Q25						

Signature of the Faculty:

Total Marks:

QUESTIONS: 25

Total Marks:25

1. **What is an Autoencoder in Deep Learning?**

ANSWER: An Autoencoder is a neural network that learns to encode input data into a compact representation and reconstruct the original data.

2. **What is the main purpose of an Autoencoder?**

ANSWER: Its main purpose is to learn useful features and reconstruct input data with minimum error.

3. **What are the major components of an Autoencoder?**

ANSWER: The major components are the **Encoder, Bottleneck (Latent Space), and Decoder.**

4. **What is the role of the Encoder in an Autoencoder?**

ANSWER: The Encoder converts the input data into a smaller and meaningful latent representation.

5. **What is the role of the Decoder in an Autoencoder?**

ANSWER: The Decoder reconstructs the original input from the latent representation.

6. **What is meant by the Bottleneck layer in an Autoencoder? –**

ANSWER: The Bottleneck is the compressed latent layer that stores the most important features of the input.

7. **What is meant by Reconstruction in an Autoencoder?**

ANSWER: Reconstruction is the process of generating an output that is as close as possible to the original input.

8. **How does an Autoencoder learn to reconstruct its input?**

ANSWER: It compares the reconstructed output with the original input and minimizes the reconstruction loss during training.

9. **What is the difference between an Autoencoder and a traditional Neural Network?**

ANSWER: An Autoencoder learns to reproduce its input, while a traditional neural network usually learns to predict a target output.

10. **What type of loss function is commonly used to train an Autoencoder?**
ANSWER: Common loss functions include Mean Squared Error (MSE) and Binary Cross-Entropy (BCE).
11. **Explain the basic architecture of an Autoencoder with a suitable example.**
ANSWER: For example, an image passes through **Encoder** → **Bottleneck** → **Decoder** → **Reconstructed Image**.
12. **Why is the dimensionality of the bottleneck layer usually smaller than the input layer?**
ANSWER: It forces the Autoencoder to learn and retain only the most important features.
13. **What happens if the bottleneck layer has too many neurons?**
ANSWER: The Autoencoder may simply copy the input without learning meaningful features.
14. **What happens if the bottleneck layer has very few neurons?**
ANSWER: Too much information may be lost, resulting in poor reconstruction.
15. **What is a Denoising Autoencoder?**
ANSWER: A Denoising Autoencoder learns to reconstruct clean data from a deliberately corrupted or noisy input.
16. **How can an Autoencoder be used for image denoising?**
ANSWER: A noisy image is given as input, and the Autoencoder learns to produce a clean version as output.
17. **How can Autoencoders be used for dimensionality reduction?** – ANSWER: The Encoder transforms high-dimensional data into a smaller latent representation that preserves important information.
18. **How can an Autoencoder be used for anomaly detection?**
ANSWER: It is trained on normal data, and unusual data is detected when it produces a high reconstruction error.
19. **What is the difference between a Denoising Autoencoder and a basic Autoencoder?** – ANSWER: A basic Autoencoder reconstructs the original input, while a Denoising Autoencoder reconstructs clean data from noisy input.
20. **Give some real-world applications of Autoencoders.** –
ANSWER: Applications include **image denoising, anomaly detection, dimensionality reduction, data compression, and recommendation systems**.
21. **What is a Deep Generative Model?**
ANSWER: A Deep Generative Model is a deep learning model that learns the data distribution and generates new data similar to the training data.
22. **What is the difference between a generative model and a discriminative model?** –
ANSWER: A generative model creates new data, while a discriminative model learns to classify or predict existing data.
23. **What is a Variational Autoencoder (VAE)?**
ANSWER: A VAE is a generative Autoencoder that learns a probability distribution in the latent space to generate new data.
24. **What is the difference between a traditional Autoencoder and a Variational Autoencoder (VAE)?**
ANSWER: A traditional Autoencoder learns fixed latent representations, while a VAE learns probabilistic latent distributions for generation.
25. **What are Generative Adversarial Networks (GANs), and what are the roles of the Generator and Discriminator?**
ANSWER: GANs consist of a **Generator** that creates fake data and a **Discriminator** that distinguishes real data from generated data.

PHOTO:**RUBRICS FOR CALCULATION:**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Subject Knowledge and Conceptual Understanding	30%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Technical Accuracy and Application of Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Problem-Solving and Analytical Thinking	20%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Communication Skills and Clarity of Explanation	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Confidence, Presentation	10%	Thorough analysis with validated	Good analysis with reasonable	Basic analysis with limited validation	Poor or no analysis